

15.3

Calculate the price of a three-month European put option on a non-dividend-paying stock with a strike price of \$50 when the current stock price is \$50, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum.

We use the BSM formulas, with $S_0 = 50, K = 50, r = 0.10, \sigma = 0.30, T = \frac{3}{12} = \frac{1}{4}$:

$$\begin{aligned} d_1 &= \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \\ &= \frac{\ln(\frac{50}{50}) + (0.10 + \frac{0.3^2}{2}) \cdot \frac{1}{4}}{0.3 \cdot \sqrt{\frac{1}{4}}} \\ d_1 &= \frac{0 + 0.03625}{0.15} = 0.2416666666 \end{aligned}$$

and accordingly:

$$\begin{aligned} d_2 &= d_1 - \sigma\sqrt{T} \\ d_2 &= 0.2416666666 - 0.3 \cdot \sqrt{\frac{1}{4}} = 0.2416666666 - 0.15 = 0.0916666666 \end{aligned}$$

The price of the put is therefore

$$\begin{aligned} p &= Ke^{-rT}N(-d_2) - S_0N(-d_1) \\ &= 50e^{-0.1 \cdot 0.25}N(-0.0916666666) - 50N(-0.2416666666) \end{aligned}$$

looking at a standard normal cdf table and approximating:

$$\begin{aligned} &= 50e^{-0.025} \cdot 0.4641 - 50 \cdot 0.4052 \\ p &= 22.6320665084 - 20.26 = 2.3720665084 \approx \boxed{\$2.37} \end{aligned}$$

15.4

What difference does it make to your calculations in Problem 15.3 if a dividend of \$1.50 is expected in two months?

The addition of the dividend means that we have to discount the current stock price accordingly:

$$S_0 = 50 - 1.50e^{-0.10 \cdot \frac{2}{12}} = 50 - 1.50e^{-\frac{1}{60}}$$

$$S_0 = 50 - 1.50 \cdot 0.98347145382 = 50 - 1.47520718073 = 48.5247928193$$

We now plug in this new value for S_0 , keeping the other values the same:

$$\begin{aligned} d_1 &= \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \\ &= \frac{\ln(\frac{48.5247928193}{50}) + (0.10 + \frac{0.3^2}{2}) \cdot \frac{1}{4}}{0.3 \cdot \sqrt{\frac{1}{4}}} \\ d_1 &= \frac{-0.02994814595 + 0.03625}{0.15} = 0.04201236033 \end{aligned}$$

and accordingly:

$$\begin{aligned} d_2 &= d_1 - \sigma\sqrt{T} \\ d_2 &= 0.04201236033 - 0.3 \cdot \sqrt{\frac{1}{4}} = 0.04201236033 - 0.15 = -0.10798763967 \end{aligned}$$

The price of the put is therefore

$$\begin{aligned} p &= Ke^{-rT}N(-d_2) - S_0N(-d_1) \\ &= 50e^{-0.1 \cdot 0.25}N(0.10798763967) - 50N(-0.04201236033) \end{aligned}$$

looking at a standard normal cdf table and approximating:

$$\begin{aligned} &= 50e^{-0.025} \cdot 0.5438 - 48.5247928193 \cdot 0.4840 \\ p &= 26.5186765078 - 23.4859997245 = 3.0326767833 \approx \boxed{\$3.03} \end{aligned}$$

15.9

Assume that a non-dividend-paying stock has an expected return of μ and a volatility of σ . An innovative financial institution has just announced that it will trade a derivative that pays off a dollar amount equal to $\ln S_T$ at time T where S_T denotes the values of the stock price at time T .

(a)

Use risk-neutral valuation to calculate the price of the derivative at time t in term of the stock price, S , at time t

To use the risk-neutral valuation, we compute the expected value assuming that the stock's expected return is the risk-free rate r , and then discount accordingly. From the lognormal property of stock prices, we know that

$$\ln(S_T) \sim \phi[\ln(S_0) + (\mu - \frac{\sigma^2}{2})T, \sigma^2 T]$$

so the risk-neutral expected value is

$$\ln(S) + (r - \frac{\sigma^2}{2})(T - t)$$

and discounting accordingly, the price of the derivative should be

$$\boxed{[\ln(S) + (r - \frac{\sigma^2}{2})(T - t)]e^{-r(T-t)}}$$

(b)

Confirm that your price satisfies the differential equation (15.16)

Equation (15.16) indicates that

$$\frac{\partial f}{\partial t} + rS \frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

and with $f = [\ln(S) + (r - \frac{\sigma^2}{2})(T - t)]e^{-r(T-t)}$, we can calculate the following:

$$\begin{aligned} - \frac{\partial f}{\partial t} &= [\ln(S) + (r - \frac{\sigma^2}{2})(T - t)] \cdot re^{-r(T-t)} - [r - \frac{\sigma^2}{2}] \cdot e^{-r(T-t)} \\ - \frac{\partial f}{\partial S} &= (\frac{1}{S} + 0)e^{-r(T-t)} = \frac{e^{-r(T-t)}}{S} \\ - \frac{\partial^2 f}{\partial S^2} &= -\frac{e^{-r(T-t)}}{S^2} \end{aligned}$$

So plugging this in:

$$\begin{aligned} &[\ln(S) + (r - \frac{\sigma^2}{2})(T - t)]re^{-r(T-t)} - [r - \frac{\sigma^2}{2}]e^{-r(T-t)} + rS \cdot \frac{e^{-r(T-t)}}{S} + \frac{1}{2}\sigma^2 S^2 \cdot -\frac{e^{-r(T-t)}}{S^2} \\ &= e^{-r(T-t)}[r \ln(S) + r(r - \frac{\sigma^2}{2})(T - t) - r + \frac{\sigma^2}{2} + r - \frac{\sigma^2}{2}] \end{aligned}$$

$$\begin{aligned}
&= e^{-r(T-t)} \left[r \ln(S) + r \left(r - \frac{\sigma^2}{2} \right) (T-t) \right] \\
&= r \cdot \left[\ln(S) + \left(r - \frac{\sigma^2}{2} \right) (T-t) \right] e^{-r(T-t)} \\
&= rf
\end{aligned}$$

exactly as we aimed to find.

15.11

What is the price of a European call option on a non-dividend-paying stock when the stock price is \$52, the strike price is \$50, the risk-free interest rate is 12% per annum, the volatility is 30% per annum, and the time to maturity is three months?

We use the BSM formulas, with $S_0 = 52$, $K = 50$, $r = 0.12$, $\sigma = 0.30$, $T = \frac{3}{12} = \frac{1}{4}$:

$$\begin{aligned}
d_1 &= \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \\
&= \frac{\ln(\frac{52}{50}) + (0.12 + \frac{0.3^2}{2}) \cdot \frac{1}{4}}{0.3 \cdot \sqrt{\frac{1}{4}}} \\
d_1 &= \frac{0.03922071315 + 0.04125}{0.15} = \frac{0.08047071315}{0.15} = 0.536471421
\end{aligned}$$

and accordingly:

$$\begin{aligned}
d_2 &= d_1 - \sigma\sqrt{T} \\
d_2 &= 0.536471421 - 0.3 \cdot \sqrt{\frac{1}{4}} = 0.536471421 - 0.15 = 0.386471421
\end{aligned}$$

The price of the call is therefore

$$\begin{aligned}
c &= S_0 N(d_1) - Ke^{-rT} N(d_2) \\
&= 52N(0.536471421) - 50e^{-0.12 \cdot 0.25} N(0.386471421)
\end{aligned}$$

looking at a standard normal cdf table and approximating:

$$= 52 \cdot 0.7054 - 50e^{-0.03} \cdot 0.6517$$

$$c = 36.6808 - 31.6219677104 = 5.0588322896 \approx \boxed{\$5.06}$$

15.13

Consider an American call option on a stock. The stock price is \$70, the time to maturity is eight months, the risk-free rate of interest is 10% per annum, the exercise price is \$65, and the volatility is 32%. A dividend of \$1 is expected after three months and again after six months. Show that it can never be optimal to exercise the option on either of the two dividend dates. Use DerivaGem to calculate the price of the option.

For the American call option, we have $S_0 = 70, T = \frac{8}{12} = \frac{2}{3}, r = 0.10, K = 65, \sigma = 0.32, D_1 = D_2 = 1$. We know that if

$$D_n \leq K[1 - e^{-r(T-t_n)}]$$

it cannot be optimal to exercise at time t_n . We want to show that it can never be optimal to exercise the option on either of the two dividend dates, t_1 and t_2 .

Evaluating first for t_1 :

$$K[1 - e^{-r(T-t_n)}]$$

where $T = \frac{6}{12}$ as the next relevant date is the next dividend date, and $t_n = \frac{3}{12}$ as normal:

$$\begin{aligned} & 65 \cdot [1 - e^{-0.1 \cdot (\frac{6}{12} - \frac{3}{12})}] \\ &= 65 \cdot [1 - e^{-0.1 \cdot \frac{1}{4}}] = 65 \cdot 0.02469008798 = \boxed{1.6048557187 \geq D_1 = 1} \end{aligned}$$

So at t_1 it is indeed not optimal to exercise. Checking now for t_2 :

$$\begin{aligned} & 65 \cdot [1 - e^{-0.1 \cdot (\frac{8}{12} - \frac{6}{12})}] \\ &= 65 \cdot [1 - e^{-0.1 \cdot \frac{1}{6}}] = 65 \cdot 0.01652854618 = \boxed{1.0743555017 \geq D_2 = 1} \end{aligned}$$

So it is also not optimal to exercise at t_2 , as we sought to find.

Finally, using DerivaGem to calculate the price of the option:

Underlying Type:		Time (Yrs)	Dividend
Equity		0.25	1
		0.5	1

Stock Price:	70.00
Volatility (% per year):	32.00%
Risk-Free Rate (% per year):	10.00%

Option Type:		☐ Imply Volatility
Binomial: American		☐ Put
Life (Years):	0.6667	☒ Call
Strike Price:	65.00	
Tree Steps:	100	

Calculate

Display Tree

Results:

Price:	10.9385523
Delta (per \$):	0.71290962
Gamma (per \$ per \$):	0.01927272
Vega (per %):	0
Theta (per day):	-0.02283062
Rho (per %):	0.25672345

We find the price to be approximately **\$10.94**.

15.15

With the notation used in this chapter

(a)

What is $N'(x)$?

$N(x)$ represents the cumulative probability that that a variable within a standard-normal distribution will be less than x , i.e. the cdf, $N'(x)$ is the derivative of the cdf, i.e. the pdf or just the probability density for the variable being equal to x . More explicitly, this would just be the function $\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$ which is very familiar by now.

(b)

Show that $SN'(d_1) = Ke^{-r(T-t)}N'(d_2)$, where S is the stock price at time t

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}$$
$$d_2 = \frac{\ln(S/K) + (r - \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}$$

Since $d_1 = d_2 + \sigma\sqrt{T - t}$, we know that

$$\begin{aligned} N'(d_1) &= N'(d_2 + \sigma\sqrt{T - t}) \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{d_2^2}{2} - \sigma d_2 \sqrt{T - t} - \frac{1}{2}\sigma^2(T - t)} \\ &= N'(d_2) e^{-\sigma d_2 \sqrt{T - t} - \frac{1}{2}\sigma^2(T - t)} \end{aligned}$$

Plugging in the given values for d_2 :

$$\begin{aligned} &= N'(d_2) e^{-\sigma\sqrt{T-t} \cdot \frac{\ln(S/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} - \frac{1}{2}\sigma^2(T-t)} \\ &= N'(d_2) e^{-\ln(S/K) - (r - \sigma^2/2)(T-t) - \frac{1}{2}\sigma^2(T-t)} \\ &= N'(d_2) e^{-\ln(S/K) - r(T-t)} \\ N'(d_1) &= N'(d_2) \frac{K}{S} e^{-r(T-t)} \\ SN'(d_1) &= Ke^{-r(T-t)} N'(d_2) \end{aligned}$$

which is precisely what we sought to find.

(c)

Calculate $\partial d_1 / \partial S$ and $\partial d_2 / \partial S$.

For d_1 :

$$\begin{aligned} d_1 &= \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \\ \frac{\partial d_1}{\partial S} &= \frac{\partial}{\partial S} \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \\ &= \frac{\partial}{\partial S} \frac{\ln(S) - \ln(K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \end{aligned}$$

$$\frac{\partial d_1}{\partial S} = \frac{1 - 0 + 0}{S\sigma\sqrt{T-t}} = \boxed{\frac{1}{S\sigma\sqrt{T-t}}}$$

For d_2 :

$$\begin{aligned} d_2 &= \frac{\ln(S/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ \frac{\partial d_2}{\partial S} &= \frac{\partial}{\partial S} \frac{\ln(S/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ &= \frac{\partial}{\partial S} \frac{\ln(S) - \ln(K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ \frac{\partial d_2}{\partial S} &= \frac{1 - 0 + 0}{S\sigma\sqrt{T-t}} = \boxed{\frac{1}{S\sigma\sqrt{T-t}}} \end{aligned}$$

(d)

Show that when

$$\begin{aligned} c &= SN(d_1) - Ke^{-r(T-t)}N(d_2) \\ \frac{\partial c}{\partial d} &= -rKe^{-r(T-t)}N(d_2) - SN'(d_1)\frac{\sigma}{2\sqrt{T-t}} \end{aligned}$$

We start with

$$c = SN(d_1) - Ke^{-r(T-t)}N(d_2)$$

Differentiating:

$$\frac{\partial c}{\partial t} = SN'(d_1)\frac{\partial d_1}{\partial t} - Ke^{-r(T-t)} \cdot r \cdot N(d_2) - Ke^{-r(T-t)} \cdot N'(d_2)\frac{\partial d_2}{\partial t}$$

We previously showed that $SN'(d_1) = Ke^{-r(T-t)}N'(d_2)$, so substitute accordingly:

$$\begin{aligned} &= SN'(d_1)\frac{\partial d_1}{\partial t} - rKe^{-r(T-t)}N(d_2) - SN'(d_1)\frac{\partial d_2}{\partial t} \\ &= SN'(d_1)\left(\frac{\partial d_1}{\partial t} - \frac{\partial d_2}{\partial t}\right) - rKe^{-r(T-t)}N(d_2) \end{aligned}$$

Looking at the first term, we can derive this partial derivative as well; we know that

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

$$\sigma\sqrt{T-t} = d_1 - d_2$$

In our case, $T = T - t$. Now differentiating:

$$\frac{\partial}{\partial t}\sigma\sqrt{T-t} = \frac{\partial d_1}{\partial t} - \frac{\partial d_2}{\partial t}$$

$$\begin{aligned}\frac{\partial d_1}{\partial t} - \frac{\partial d_2}{\partial t} &= \sigma \cdot (T - t)^{-\frac{1}{2}} \cdot -\frac{1}{2} \\ \frac{\partial d_1}{\partial t} - \frac{\partial d_2}{\partial t} &= -\frac{\sigma}{2\sqrt{T-t}}\end{aligned}$$

Finally, plugging this in:

$$\boxed{\frac{\partial \mathbf{c}}{\partial \mathbf{t}} = -\frac{\mathbf{S}\mathbf{N}'(\mathbf{d}_1)\sigma}{2\sqrt{\mathbf{T}-\mathbf{t}}} - \mathbf{r}\mathbf{K}\mathbf{e}^{-\mathbf{r}(\mathbf{T}-\mathbf{t})}\mathbf{N}(\mathbf{d}_2)}$$

(e)

Show that $\partial c / \partial S = N(d_1)$.

We start with

$$c = SN(d_1) - Ke^{-r(T-t)}N(d_2)$$

Differentiating:

$$\frac{\partial c}{\partial S} = N(d_1) + SN'(d_1)\frac{\partial d_1}{\partial S} - Ke^{-r(T-t)}N'(d_2)\frac{\partial d_2}{\partial S}$$

We previously showed that $SN'(d_1) = Ke^{-r(T-t)}N'(d_2)$, so substitute accordingly:

$$\begin{aligned}\frac{\partial c}{\partial S} &= N(d_1) + SN'(d_1)\frac{\partial d_1}{\partial S} - SN'(d_1)\frac{\partial d_2}{\partial S} \\ &= N(d_1) + SN'(d_1)\left(\frac{\partial d_1}{\partial S} - \frac{\partial d_2}{\partial S}\right)\end{aligned}$$

We also previously showed that $\frac{\partial d_1}{\partial S} = \frac{\partial d_2}{\partial S} = \frac{1}{S\sigma\sqrt{T-t}}$, so substitute accordingly:

$$= N(d_1) + SN'(d_1)\left(\frac{1}{S\sigma\sqrt{T-t}} - \frac{1}{S\sigma\sqrt{T-t}}\right)$$

$$\boxed{\frac{\partial \mathbf{c}}{\partial \mathbf{S}} = \mathbf{N}(\mathbf{d}_1)}$$

(f)

Show that c satisfies the Black-Scholes-Merton differential equation.

We aim to show that

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

where $f = c$. We've already found $\frac{\partial c}{\partial t}$ and $\frac{\partial c}{\partial S}$, let's find $\frac{\partial^2 c}{\partial S^2}$:

$$\begin{aligned}\frac{\partial^2 c}{\partial S^2} &= \frac{\partial}{\partial S} N(d_1) \\ \frac{\partial^2 c}{\partial S^2} &= N'(d_1) \frac{\partial d_1}{\partial S} \\ \frac{\partial^2 c}{\partial S^2} &= \frac{N'(d_1)}{S\sigma\sqrt{T-t}}\end{aligned}$$

Let's now substitute accordingly:

$$\begin{aligned}& \frac{\partial c}{\partial t} + rS \frac{\partial c}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 c}{\partial S^2} \\ &= -\frac{SN'(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}N(d_2) + rSN(d_1) + \frac{1}{2}\sigma^2 S^2 \cdot \frac{N'(d_1)}{S\sigma\sqrt{T-t}} \\ &= -\frac{SN'(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}N(d_2) + rSN(d_1) + \frac{SN'(d_1)\sigma}{2\sqrt{T-t}} \\ &= -rKe^{-r(T-t)}N(d_2) + rSN(d_1) \\ &= r(SN(d_1) - Ke^{-r(T-t)}N(d_2)) \\ &= rc\end{aligned}$$

Exactly as we sought to find.

(g)

Show that c satisfies the boundary condition for a European call option, i.e., that $c = \max(S - K, 0)$ as t tends to T .

We know that $c = SN(d_1) - Ke^{-r(T-t)}N(d_2)$. As t approaches T :

- If $S > K$: We have $\ln(\frac{S}{K}) > 0$ while $\sigma\sqrt{T-t}$ approaches 0, so d_1 and d_2 approach infinity. Therefore $N(d_1)$ and $N(d_2)$ approach 1. c essentially becomes $S - K$, which given our initial assumption, is positive and therefore greater than 0. **The condition holds**.

- If $S < K$: We have $\ln(\frac{S}{K}) < 0$ while $\sigma\sqrt{T-t}$ approaches 0, so d_1 and d_2 approach negative infinity. Therefore $N(d_1)$ and $N(d_2)$ approach 0. c therefore basically just becomes 0, which given our initial assumption, would be greater than $S - K$, which would be negative, so **the condition also holds**.

15.26

Suppose that observations on a stock price (in dollars) at the end of each of 15 consecutive weeks are as follows:

30.2, 32.0, 31.1, 30.1, 30.2, 30.3, 30.6, 33.0, 32.9, 33.0, 33.5, 33.5, 33.7, 33.5, 33.2

Estimate the stock price volatility. What is the standard error of your estimate?

To calculate the stock price volatility, we take the weekly log returns $u_i = \ln(\frac{S_i}{S_{i-1}})$. We can then take the weekly standard deviation and annualize to 52 weeks, and then take the standard error accordingly. I did this in python to make it easier:

```
import numpy as np

weekly_returns = np.array([30.2, 32.0, 31.1, 30.1, 30.2, 30.3, 30.6, 33.0, 32.9, 33.0, 33.5, 33.5, 33.7, 33.5, 33.2])

u = np.log(weekly_returns[1:] / weekly_returns[:-1])
sum_u = np.sum(u)
sum_u_squared = np.sum(u**2)

weekly_returns_sd = np.sqrt((sum_u_squared - (sum_u**2) / len(weekly_returns)))

annual_vol = weekly_returns_sd * np.sqrt(52)

vol_standard_error = annual_vol / np.sqrt(2 * len(weekly_returns))

print(f"Weekly returns SD: {weekly_returns_sd}")
print(f"Annual volatility estimate: {annual_vol}")
print(f"Standard error of estimate: {vol_standard_error}")
```

with the following output:

```
> python 15_26.py
Weekly returns SD: 0.027841997134013863
```

Annual volatility estimate: 0.20077149655601684

Standard error of estimate: 0.0366556925226004

So we see that the annual volatility estimate is approximately **20.08%** and the standard error of that estimate is approximately **3.67%**.